

2.1/2.2 Restricting, Simplifying, Multiplying, and Dividing Rational Expressions

Lesson Outline:

Part 1: Stating restrictions

Part 2: Simplifying rational expressions

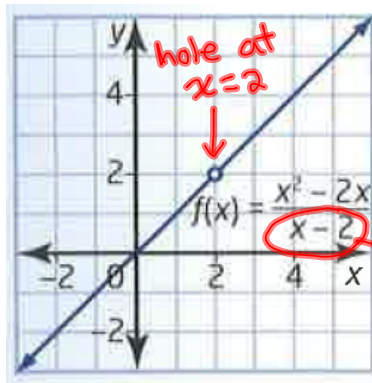
Part 3: Multiplying rational expressions

Part 4: Dividing rational expressions

What is a rational expression?

Rational expression: the quotient of two polynomials, $\frac{p(x)}{q(x)}$, where $q(x) \neq 0$.

Example of a graph of a rational expression:



The open circle is used to represent a hole in the graph. This corresponds to any restrictions on the variable (denominator can't be 0).

$$x - 2 \neq 0$$
$$x \neq 2$$

Stating Restrictions

Note: rational expressions must be checked for restrictions by determining where the denominator is equal to **zero**. These restrictions must be stated when the expression is simplified.

bottom of a fraction can
NOT = 0.

Example 1: State the restrictions for the following rational expressions

a)

$$\frac{x+2}{x-2}$$

$$x-2 \neq 0$$
$$x \neq 2$$

$$x \neq 2$$

b)

$$\frac{x+2}{(x-3)(x+4)}$$

$$x-3 \neq 0$$
$$x \neq 3$$

$$x+4 \neq 0$$
$$x \neq -4$$

$$x \neq 3, -4$$

c)

$$\frac{5}{x(x+3)}$$

$$x \neq 0$$

$$x+3 \neq 0$$

$$x \neq -3$$

$$x \neq 0, -3$$

Rule: We can cancel out **ONLY** when multiplying fractions

You can cancel factors.

e.g. $\frac{\cancel{x}(x+1)}{\cancel{x}} = x+1$

e.g. $\frac{3\cancel{x}(x+3)}{2\cancel{4}x} = \frac{3x+9}{2}$

These are examples of when cancelling is allowed.

Rule: We can **NOT** cancel out when adding or subtracting fractions

e.g. $\frac{\cancel{x}+8}{\cancel{x}}$ **NO!** Don't do this!

e.g. $\frac{2x+3}{4x}$ No cancelling!

Simplifying Rational Expressions

Example 2: Simplifying each expression and determine any restrictions on the variable.

a) $\frac{3x^2}{y\cancel{x}}$, $y \neq 0$
 $x \neq 0$

$$= \frac{3\cancel{x}}{y} , \quad y \neq 0$$
$$x \neq 0$$

b) $\frac{x-3}{x^2+3x-18}$

Note: factor where possible and then state restrictions before cancelling factors.

$$= \frac{\cancel{x-3}}{(x+6)(\cancel{x-3})} , \quad x \neq -6, 3$$

$$= \frac{1}{x+6} , \quad x \neq -6, 3$$

c)

$$\frac{x^2 + 10x + 21}{x + 3} \quad , x \neq -3$$

$$= \frac{(x+7)\cancel{(x+3)}}{\cancel{x+3}} \quad , x \neq -3$$

$$= x + 7 \quad , x \neq -3$$

d)

$$\frac{x + 1}{x^2 + 3x + 2}$$

$$= \frac{\cancel{x+1}}{(x+2)\cancel{(x+1)}} \quad , x \neq -2, -1$$

$$= \frac{1}{x+2}$$

e)

$$\frac{x^2 - 9}{x^2 + 7x + 12} \leftarrow \text{difference of squares: } a^2 - b^2 = (a-b)(a+b)$$

$$= \frac{(x-3)(\cancel{x+3})}{(x+4)(\cancel{x+3})} \quad , x \neq -4, -3$$

$$= \frac{x-3}{x+4} \quad , x \neq -4, -3$$

f)

$$\frac{6x^2 - 7x - 5 *}{3x^2 + x - 10 *}$$

factor numerator

$$\begin{aligned} &6x^2 - 10x + 3x - 5 \\ &= (6x^2 - 10x) + (3x - 5) \\ &= 2x(3x - 5) + 1(3x - 5) \\ &= (2x+1)(3x-5) \end{aligned}$$

factor denominator

$$\begin{aligned} &3x^2 + x - 10 \\ &= 3x^2 + 6x - 5x - 10 \\ &= (3x^2 + 6x) + (-5x - 10) \\ &= 3x(x+2) - 5(x+2) \\ &= (x+2)(3x-5) \end{aligned}$$

$$= \frac{(2x+1)(\cancel{3x-5})}{(x+2)(\cancel{3x-5})} \quad , x \neq -2, \frac{5}{3}$$

$$= \frac{2x+1}{x+2} \quad , x \neq -2, \frac{5}{3}$$

Multiplying Rational Expressions

a)

$$\frac{2\cancel{4}x^{\cancel{2}^1}}{1\cancel{3}x} \cdot \frac{4\cancel{12}x^{\cancel{3}^2}}{1\cancel{2}x} ; x \neq 0$$

1. factor where possible
2. cancel common factors
3. multiply numerators and denominators
4. state restrictions (throughout process)

$$= 8x^3 ; x \neq 0$$

b)

$$\frac{4x+24}{x^2+8x} \cdot \frac{12x^2}{3x+18}$$

$$= \frac{4(\cancel{x+6})}{x(\cancel{x+8})} \cdot \frac{4\cancel{12}x^{\cancel{2}^1}}{1\cancel{3}(\cancel{x+6})} ; x \neq 0, -8, -6$$

$$= \frac{16x}{x+8} ; x \neq 0, -8, -6$$

c)

$$\frac{x+1}{2x} \cdot \frac{3x}{x^2+4x+3} \quad , x \neq 0$$

$$= \frac{\cancel{x+1}}{2\cancel{x}} \cdot \frac{3\cancel{x}}{(\cancel{x+1})(x+3)} \quad ; x \neq 0, -1, -3$$

$$= \frac{3}{2(x+3)} \quad ; x \neq 0, -1, -3$$

d)

$$\frac{*5x^2-13x+8}{x-7} \cdot \frac{1}{5x-8}$$

$$\begin{aligned} &\text{factor} \\ &= 5x^2 - 8x - 5x + 8 \\ &= (5x^2 - 8x) + (-5x + 8) \\ &= x(5x - 8) - 1(5x - 8) \\ &= (5x - 8)(x - 1) \end{aligned}$$

$$= \frac{(\cancel{5x-8})(x-1)}{x-7} \cdot \frac{1}{\cancel{5x-8}} \quad ; x \neq 7, \frac{8}{5}$$

$$= \frac{x-1}{x-7} \quad ; x \neq 7, \frac{8}{5}$$

Dividing Rational Expressions

a)

$$\frac{10ab^2}{4a} \div \frac{15a^2}{12b^2}$$

*no cross cancelling until
after second fraction has
been flipped*

1. flip second fraction and change to multiplication
2. factor where possible
3. cancel common factors
4. multiply numerators and denominators
5. state restrictions (throughout process)

$$= \frac{10ab^2}{4a} \times \frac{12b^2}{15a^2}$$

$$; a \neq 0$$

$$b \neq 0$$

*because 'b' was
in the denominator of
the original expression*

$$= \frac{\cancel{2} \cancel{12} \cancel{a} b^4}{\cancel{4} \cancel{a} \cancel{15} a^2}$$

$$; a \neq 0$$

$$b \neq 0$$

$$= \frac{2b^4}{a^2} ; a \neq 0$$

$$b \neq 0$$

b)

$$\frac{a^2 + 2a}{3a} \div \frac{5a^2 + 10a}{20a^2}$$

$$= \frac{a^2 + 2a}{3a} \times \frac{20a^2}{5a^2 + 10a} ; a \neq 0$$

$$= \frac{\cancel{a}(\cancel{a} + 2)}{\cancel{3}\cancel{a}} \times \frac{\cancel{4} \cancel{20} a^2}{\cancel{5} a(\cancel{a} + 2)} ; a \neq 0, -2$$

$$= \frac{4a}{3} ; a \neq 0, -2$$

c)

$$\frac{2x^2 - 8x}{x^2 - 3x - 10} \div \frac{4x^2}{x^2 - 9x + 20}$$

$$= \frac{2x^2 - 8x}{x^2 - 3x - 10} \times \frac{x^2 - 9x + 20}{4x^2} ; x \neq 0$$

$$= \frac{\cancel{2}x(\cancel{x}-4)}{\cancel{(x-5)}(x+2)} \times \frac{(\cancel{x-5})(x-4)}{\cancel{2}4x^{\cancel{2}}} ; x \neq 0, 5, -2, 4$$

$$= \frac{(x-4)^2}{2x(x+2)} ; x \neq 0, 5, -2, 4$$

DO WORKSHEET